

Noise sweep, T1x test (n=287): HIP reference (left) vs SCINE/DFTB0 + xTB/GFN1 (right)

Convergence rate (%)
strict: $n_{\text{neg}}=1 \wedge f_{\text{max}} < 0.01$

HIP (reference) vs SCINE/DFTB0 (dt=0.007, 15k steps): GAD matches HIP strict-conv (89.9% @10pm); GAD IRC TOPO > Sella's (+12.9pp @10pm)

